ITI – RedHat Admin 2 lab4

Mohammad Tarek Omar – Data Management

1. **Create a program or command that allows a normal user to execute it with the file owner's privileges.**

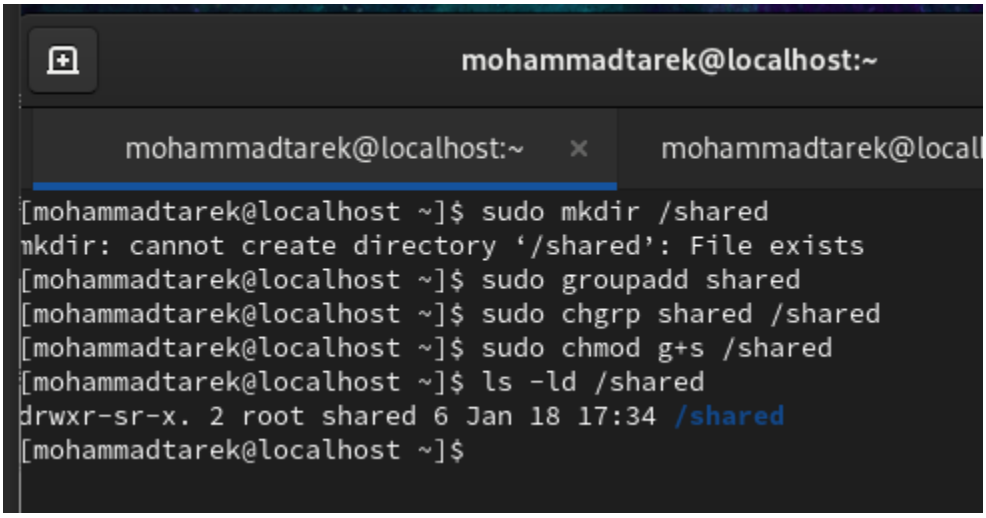
```
root@localhost:/home/mohammadtar
root@localhost:/home/mohamma...   mohammadtarek@
[mohammadtarek@localhost ~]$ sudo su root
[sudo] password for mohammadtarek:
[root@localhost mohammadtarek]# vi task.c
[root@localhost mohammadtarek]#
```

```
root@localhost:/home/mohammadtarek
root@localhost:/home/mohamma...   mohammadtarek@localhost:~ — ...
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(){
        setuid(0);
        printf(geteuid());
        system("cat /etc/shadow | head -n 1");
        return 0;
}
~
~
```

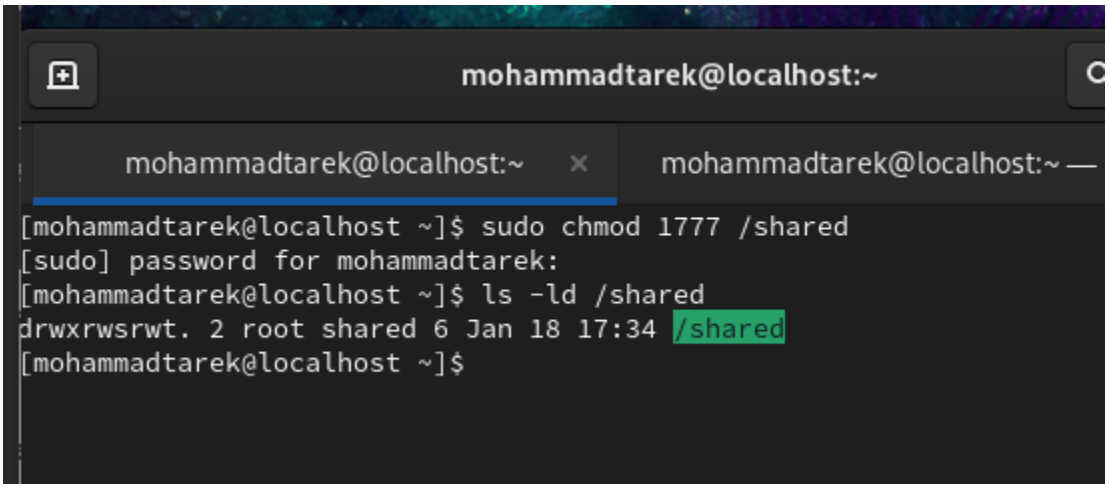
```
[root@localhost mohammadtarek]# gcc task.c -o my_task
[root@localhost mohammadtarek]# sudo chmod u+s my_task
[root@localhost mohammadtarek]# sudo su mohammadtarek
[mohammadtarek@localhost ~]$ ./my_task
root:$6$GpJlPh7XqlAtdY7h$MxMQX4EAizAytiDt/xkHPcyMirYFBsiP9dG.ARjZMNAWZ38/kzVpFk6
jTLUXl91i6AhOw93iRu2kH/bQyjlYb1::0:99999:7:::
UID[mohammadtarek@localhost ~]$ ls -l /my_task
ls: cannot access '/my_task': No such file or directory
[mohammadtarek@localhost ~]$ ls -l my_task
-rwsr-xr-x. 1 root root 17600 Jan 18 18:22 my_task
[mohammadtarek@localhost ~]$
```

2. **Why does SetUID not work on shell scripts?** SetUID is ignored on shell scripts because the kernel executes the interpreter (e.g., /bin/bash) instead of the script itself, which introduces race conditions and serious security vulnerabilities. Therefore, modern Linux kernels disable SetUID on scripts.

3. **Ensure all files created in a directory inherit the directory's group.**



4. **Configure a shared directory where users can only delete their own files.**



5. **What happens without Sticky Bit?** Without the Sticky Bit, any user who has write access to a directory can delete or rename files owned by other users because file deletion depends on directory permissions, not file ownership.

6. **Give user ali read/write access to a file without changing ownership.**

```
[mohammadtarek@localhost ~]$ sudo adduser ali
[sudo] password for mohammadtarek:
[mohammadtarek@localhost ~]$ sudo passwd ali
Changing password for user ali.
New password:
```

```
[mohammadtarek@localhost ~]$ setfacl -m u:ali:rw file22
[mohammadtarek@localhost ~]$ ls -l file22
-rwsrw-r--+ 1 mohammadtarek mohammadtarek 46 Jan 18 17:23 file22
[mohammadtarek@localhost ~]$
```

7. **View ACLs on a file data.txt**

```
[mohammadtarek@localhost ~]$ touch data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
group::r--
other::r--

[mohammadtarek@localhost ~]$
```

8. **Give user alice read/write access to data.txt**

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~   mohammadtarek@localhost:~ —
[mohammadtarek@localhost ~]$ sudo adduser alice
[mohammadtarek@localhost ~]$ sudo passwd alice
Changing password for user alice.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[mohammadtarek@localhost ~]$ setfacl -m u:alice:rw data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
user:alice:rw-
group::r--
mask::rw-
other::r--

[mohammadtarek@localhost ~]$
```

9. **Remove user alice ACL from data.txt**

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~   mohammadtarek@localhost:~ — ...
[mohammadtarek@localhost ~]$ setfacl -x u:alice data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
group::r--
mask::r--
other::r--

[mohammadtarek@localhost ~]$
```

10. **Give group developers read access to data.txt**

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~   mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ sudo groupadd developers
[sudo] password for mohammadtarek:
[mohammadtarek@localhost ~]$ setfacl -m g:developers:r data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
group::r--
group:developers:r--
mask::r--
other::r--

[mohammadtarek@localhost ~]$
```

11. **Why does the group still not have read access?** It does not, but in case, because of the 'mask'

12. **Set mask so named users/groups have only read access**

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~   mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ setfacl -m m:r data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
group::r--
group:developers:r--
mask::r--
other::r--

[mohammadtarek@localhost ~]$
```

same as I told you, but this is the right command to make all users and groups to have access to read only on the file

13. Display effective permissions

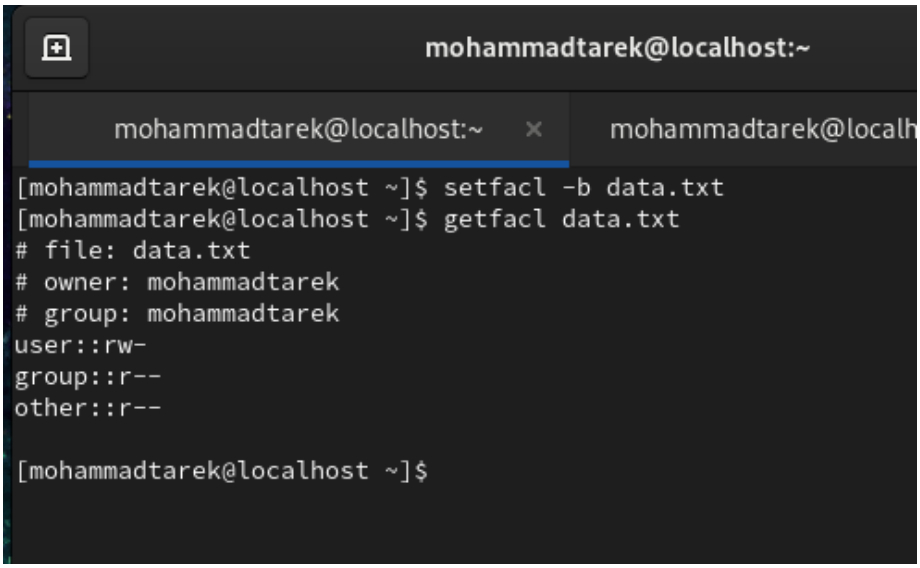
```
mohammadtarek@localhost:~
mohammadtarek@localhost:~   mohammadtarek@localhost:~
[mohammadtarek@localhost ~]$ setfacl -m g:developers:rw data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
group::r--
group:developers:rw-
mask::rw-
other::r--

[mohammadtarek@localhost ~]$ setfacl -m m:r data.txt
[mohammadtarek@localhost ~]$ getfacl data.txt
# file: data.txt
# owner: mohammadtarek
# group: mohammadtarek
user::rw-
group::r--
group:developers:rw-            #effective:r--
mask::r--
other::r--

[mohammadtarek@localhost ~]$
```

14. Apply ACLs recursively and ensure inheritance

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~    mohammadtarek@localhost:~ — ...

[mohammadtarek@localhost ~]$ vi /shared/task
[mohammadtarek@localhost ~]$ ls shared
ls: cannot access 'shared': No such file or directory
[mohammadtarek@localhost ~]$ ls /shared
task
[mohammadtarek@localhost ~]$ setfacl -R -m u:ali:rwx /shared
setfacl: /shared: Operation not permitted
[mohammadtarek@localhost ~]$ sudo setfacl -R -m u:ali:rwx /shared
[sudo] password for mohammadtarek:
[mohammadtarek@localhost ~]$ sudo setfacl -m d:u:ali:rwx /shared
[mohammadtarek@localhost ~]$ getfacl /shared
getfacl: Removing leading '/' from absolute path names
# file: shared
# owner: root
# group: shared
# flags: -st
user::rwx
user:ali:rwx
group::rwx
mask::rwx
other::rwx
default:user::rwx
default:user:ali:rwx
default:group::rwx
default:mask::rwx
```

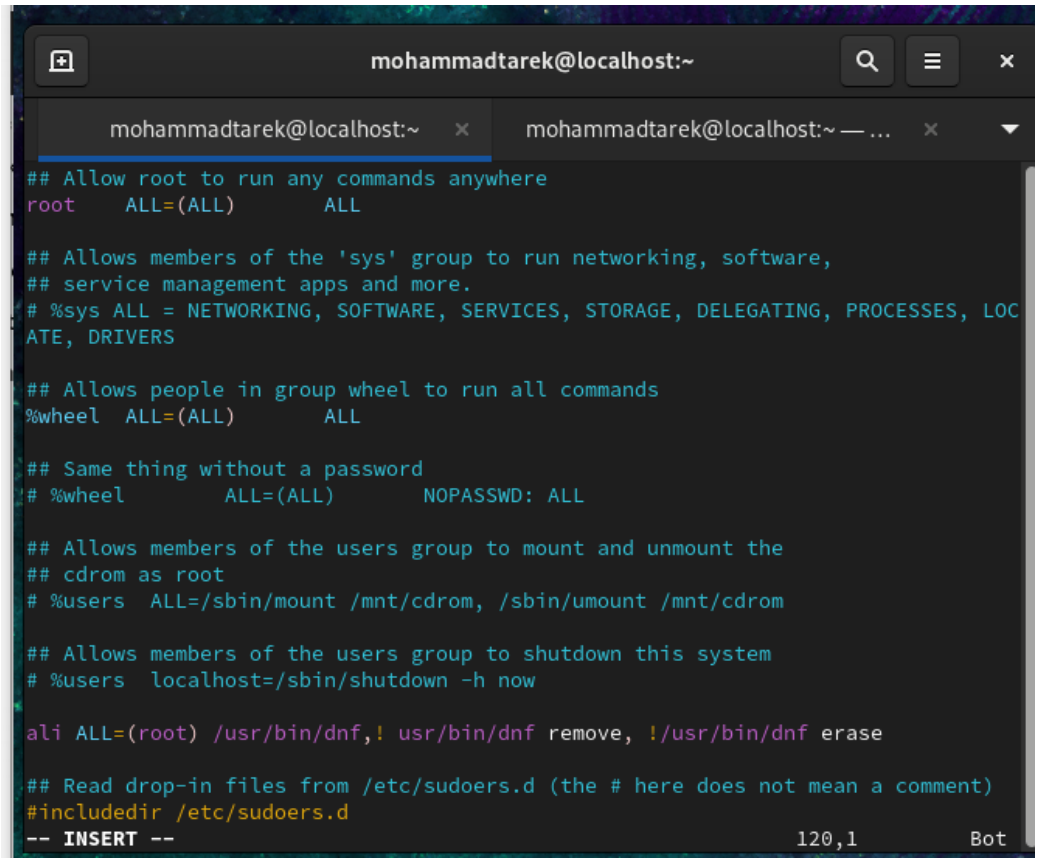
```
[mohammadtarek@localhost ~]$ getfacl /shared/task
getfacl: Removing leading '/' from absolute path names
# file: shared/task
# owner: mohammadtarek
# group: shared
user::rw-
user:ali:rwx
group::r--
mask::rwx
other::r--

[mohammadtarek@localhost ~]$
```

15. Remove all ACL entries from data.txt



16. Why should visudo be used instead of editing /etc/sudoers directly? Syntax Validation, Preventing Lockout, File Locking.

17. Allow user ali to install packages but not remove them.



6. **Allow group developers to run docker commands**

```
%developers ALL=(root) /usr/bin/docker
## Read drop-in files from /etc/sudoers.d (the # here does not mean a comment)
```

```
ali@localhost:/home/mohammadtarek

ali@localhost:/home/mohammadt...   mohammadtarek@localhost:~ — ...

[mohammadtarek@localhost ~]$ sudo usermod -aG developers ali
[mohammadtarek@localhost ~]$ sudo visudo
[mohammadtarek@localhost ~]$ sudo su ali
[ali@localhost mohammadtarek]$ sudo docker ps

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

[sudo] password for ali:
sudo: docker: command not found
[ali@localhost mohammadtarek]$
```

7. **Allow user ali to reboot the system without a password**

```
ali ALL=(ALL) NOPASSWD: /usr/sbin/reboot
```

```
ali@localhost:~

ali@localhost:~   mohamm

[mohammadtarek@localhost ~]$ sudo visudo
[sudo] password for mohammadtarek:
[mohammadtarek@localhost ~]$ su - ali
Password:
[ali@localhost ~]$ sudo reboot
[ali@localhost ~]$
```

8. **Allow group devops to run any command without a password**

```
%developers ALL=(ALL) NOPASSWD: ALL
```

```
ali@localhost:/home/mohammadtarek
[mohammadtarek@localhost ~]$ sudo visudo
[sudo] password for mohammadtarek:
/etc/sudoers:125:36: syntax error
%developers ALL=(ALL) NOPASWWD: ALL
                                   ^
What now? ^[
Options are:
  (e)dit sudoers file again
  e(x)it without saving changes to sudoers file
  (Q)uit and save changes to sudoers file (DANGER!)

What now? e
[mohammadtarek@localhost ~]$ sudo su ali
[ali@localhost mohammadtarek]$ sudo whoami
root
[ali@localhost mohammadtarek]$
```